

Monday Warm Up: Unit 4, Reactance, RL circuits, RLC circuits

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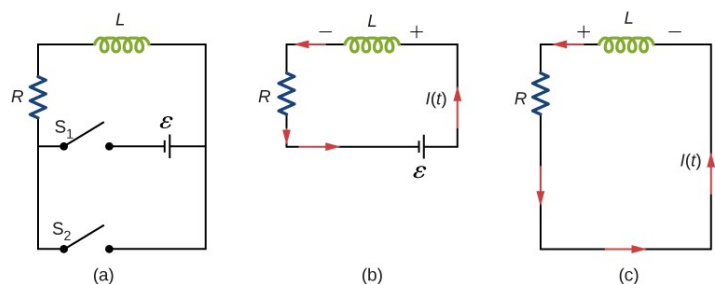


Figure 1: A circuit diagram of a DC emf with a resistor, inductor, and two switches.

1 Memory Bank

- $\tau = RC$... The time constant of an RC circuit.
- $f_0 = \frac{1}{2\pi\sqrt{LC}}$... RLC circuit resonance frequency.
- $i(t) = \frac{\mathcal{E}}{R} (1 - \exp(-t/\tau))$... Charging RL circuit.
- $i(t) = \frac{\mathcal{E}}{R} \exp(-t/\tau)$... Discharging RL circuit.
- $\tau = L/R$... Time constant of an RL circuit.
- $V_{\text{rms}} = I_{\text{rms}} Z_{\text{tot}}$... Ohm's Law for AC circuits.
- $P_{\text{ave}} = I_{\text{ave}} V_{\text{ave}} \cos \phi$... Average power in series RLC circuit.
- $X_C = 1/(2\pi fC)$... Reactance of a capacitor.
- $X_L = 2\pi fL$... Reactance of an inductor.
- $Z_{\text{tot}} = \sqrt{R^2 + (X_L - X_C)^2}$... The total impedance of an RLC series circuit.
- $\cos \phi = R/Z_{\text{tot}}$... Power factor of an RLC series circuit.

2 RL Circuits

1. Consider Fig. 1. There is a battery connected to a resistor R and an inductor L via switch 1. Switch 2 connects L and R . In algebraic terms, what are the currents when: (a) $t = 0$, when switch 1 is closed, and (b) when $t \gg \tau$?

2. (a) What is τ when $R = 1 \text{ k}\Omega$, and $L = 0.2 \text{ Henries}$? (b) What will the time constant τ be if we double R and quadruple L ?
3. If switch 1 is open and switch 2 closed after $t \gg \tau$, what time passes before the current is reduced to half of its original value?

3 Reactance and Impedance

1. (a) What is the reactance, in ohms, of a $2.5 \mu\text{F}$ capacitor at 1.4 kHz ? (b) What is the reactance of a 5.0 mH inductor at this same frequency? (c) Now add (in series) a $1 \text{ k}\Omega$ resistor. What is the total impedance at this frequency? (d) What is the resonance frequency? (e) What is the total impedance at 1.4 MHz ?
2. For an RLC series circuit having a 40.0Ω resistance, a 3.00 mH inductance, a $5.00 \mu\text{F}$ capacitance, and a voltage source with a V_{rms} of 120 V : (a) calculate the *phase angle*, ϕ , and *power factor*, $\cos \phi$, for $f = 60 \text{ Hz}$. (b) What is the average power at 60.0 Hz ? (c) Find the average power at the resonance frequency.